



Stress-resistant trees are more common in urban than rural forests: A case study of Cleveland, Ohio's natural parks

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ABSTRACT

Natural parks are comprised of preserved forested natural areas that are undergoing natural ecological processes. These areas can offer a refuge for local biodiversity and contribute substantially to ecosystem services in both rural areas with relatively low population densities, as well as high-density urban areas. Forested natural parks located in urban areas should experience more stressful environmental conditions than nearby rural areas, yet we know relatively little about how urbanization impacts tree communities within these important natural habitats. To better understand the impact of urbanization on forests, we investigated the species composition, abundance, and diversity of midstory and canopy trees as well as tree seedlings in urban and rural natural parks in and around Cleveland, Ohio. We found that both urban and rural natural parks have similar tree abundance, but midstory and canopy trees as well as tree seedling communities in the urban natural parks included higher abundances of stress-tolerant species compared to rural parks. In addition, this pattern was driven by changes in native tree species, as we observed low abundance of invasive species. More stress-resistant native species in urban areas include *Quercus rubra* and *Prunus serotina*, in contrast to rural natural parks which are dominated by *Acer* spp. and *Fagus grandifolia*. Lastly, we show that urban and rural natural parks have similar species diversity within plots, but we found higher variation in community composition among urban natural parks compared to among rural parks. Furthermore, *Q. rubra* and *P. serotina* were significantly larger in rural natural parks, indicating that both environmental stress and successional stage could drive compositional differences. Thus, we show that urbanization can have unexpected effects on plant community composition and diversity. Our study refutes the idea that these are degraded habitats, highlights the need to conserve them, and suggests that characterization of local variation in self-assembled urban tree communities will provide the most accurate picture of their management needs and potential ecosystem services.

1. Introduction

Urban areas are rapidly expanding, making the urban forest an increasingly important part of our landscape. Urban forests, which include street trees, as well as trees in parks, medians, and natural parks (Konijnendijk et al., 2006; McPherson et al., 2017), are recognized as a critical provider of ecosystem services. These ecosystem services include the regulation of local environmental conditions, e.g., carbon sequestration (Nowak et al., 2013) and evaporative cooling (Rahman et al., 2014; Scholz et al., 2018) as well as social and psychological benefits (Nesbitt et al., 2017). Because of the economic and cultural value of urban trees, urban foresters are seeking to increase the urban tree

canopy, and associated biodiversity (Alvey, 2006). As a key component of urban forests, trees in urban natural parks are important sanctuaries for local biodiversity (Crocchi et al., 2008) and store a disproportionate amount of carbon relative to other urban tree types (Yesilonis and Pouyat, 2012). While conservation of the urban forest typically focuses on the largest trees because they contribute disproportionately more to ecosystem services (Nowak and Crane, 2002), large trees are likely to have established prior to urbanization, while the future plant community composition will be determined by current patterns of seedling recruitment. Thus, fine-scale studies of the composition and diversity of both established and establishing life stages in urban forests relative to their rural counterparts, is required.

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Urban habitats are fragmented, disturbed, and present a suite of environmental characteristics including elevated CO₂ (Mitchell et al., 2018) and higher temperatures due to heat retention of impervious surfaces (Oke et al. 1989), and these are both likely to impact the structure and composition of forested areas. Thus, urban forests can be home to distinct woody plant communities, which differ both in taxonomic and functional composition compared to nearby forests in rural settings (Zipperer, 2002; Zipperer and Guntenspergen, 2009). It is difficult, however, to formulate general predictions about how urban environments impact plant community composition because spatial heterogeneity, anthropogenic impacts, and abiotic stressors may interact to have unexpected effects.

Biodiverse forests provide habitat for wildlife, increase ecosystem services, and have a greater buffer against disease and pests (Thompson et al., 2009; Isbell et al., 2011; Barth et al., 2015). Yet, a consensus on whether urbanization increases or decreases biodiversity in forests has not been reached (Kendal et al., 2014). On one hand, diversity may increase in urban forested areas as novel urban environments provide habitat for both non-native and invasive species (Williams et al., 2009). Conversely, inhospitable conditions within some urban areas may be unsuitable for sensitive native species, ultimately reducing diversity of urban tree communities (Kendal et al., 2014). Across urban areas within the US, there is evidence that urban biological communities are becoming more homogeneous, potentially lowering beta diversity in urban areas (Groffman et al., 2014).

Urban natural parks, sometimes referred to as urban remnant forests, are publicly or privately held tracts of land which have never been cleared for urbanization (Zipperer, 2002). Urban natural parks within Northeast Ohio include secondary forests and present-day land management within them is often minimal. Like their rural counterparts,

these areas are undergoing natural ecological processes like recruitment but may vary from one another in successional stage. While we are beginning to understand the impact of urbanization on the composition of urban forests broadly, there is still limited research on the variation in species composition and structure of forests undergoing natural regeneration. Moreover, the composition of the seedling layer of forested areas within this context, is often overlooked. This represents an important knowledge gap in natural parks, as the composition of woody seedlings within the forest understory provides insight into the future forest composition. Previous work comparing urban and rural seedlings has shown that urban areas generally have greater abundance of invasive seedlings (Zipperer, 2002; Cadenasso et al., 2007) and limited native species regeneration relative to rural areas (Wallace and Laughlin, 2017; Piana et al., 2021). Therefore, we need assessments of the seedling life stage to capture the dynamic community processes and predict the future of the urban forest.

To determine the impact of urbanization on natural tree communities, we explored the species composition of forests within urban and rural natural parks located in and around Cleveland, Ohio (Fig. 1). The most common forest type in the area is temperate hardwood, with *Acer saccharum*, *Acer rubrum* and *Fagus grandifolia* among the dominant species (Braun, 1950). We established 63 forested plots within 18 natural parks (Table 1) and counted midstory and canopy trees as well as seedlings to address the following questions: 1) Does tree community composition differ between urban and rural forests? 2) Do urban and rural forests differ in the abundance of midstory trees, canopy trees and seedlings? 3) Do urban forests have lower biodiversity relative to nearby rural forests? 4) Is there lower among-plot biodiversity within urban forests (i.e., lower beta-diversity) relative to rural forests?

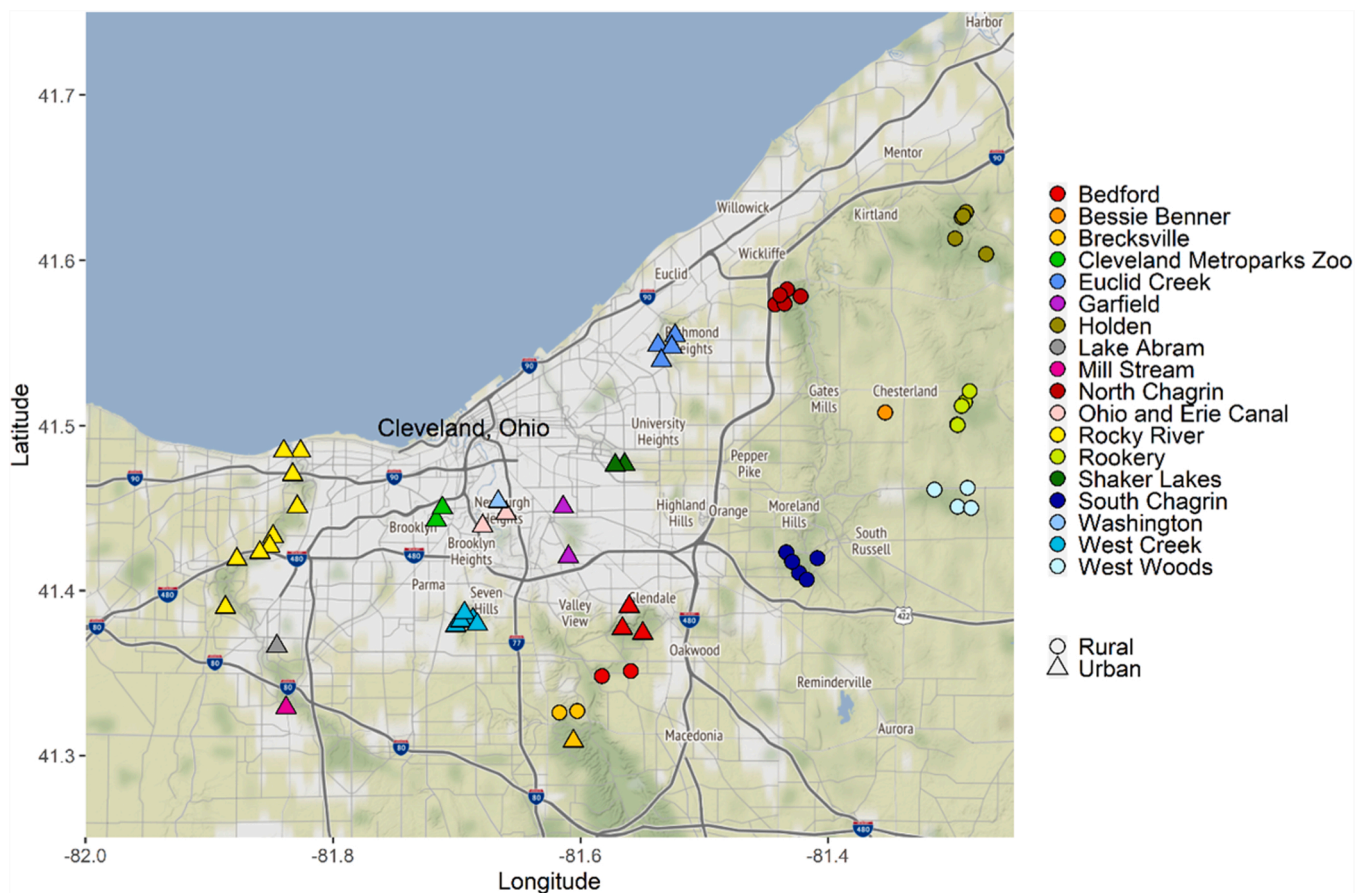


Fig. 1. Map of study area around Cleveland, Ohio, USA. Points denote urban (triangle) and rural (circle) research plots with each natural park represented in a different color. White shading indicates urban areas.

Table 1

Name of sites and number of plots sampled within natural parks) and the affiliated organizations.

Managing Entity	Natural Park Name	Number of Plots
Cleveland Metroparks	Bedford	4
	Brecksville	4
	Euclid Creek	4
	Garfield	2
	Lake Abram	1
	Mill Stream	1
	North Chagrin	4
	Ohio and Erie Canal	2
	Rocky River	9
	South Chagrin	5
	Washington	1
	West Creek	5
	Zoo	2
Geauga Park District	Bessie Benner Metzenbaum Park	1
	The Rookery	5
	The West Woods	4
The Holden Arboretum	The Holden Arboretum	5
Nature Center at Shaker Lakes	Shaker Lakes	2

2. Methods

2.1. Study area

This study took place in and around Cleveland, OH (41.4993°N, 81.6944°W), which is located along the southern border of Lake Erie. With a population of 367,991, Cleveland, OH represents a mid-size US city (US Census Bureau, 2020). The study areas spanned 201.3 km² area in northeastern Ohio. The climate is temperate with a mean annual high of 15.9 °C and a mean low temperature of 6.7 °C. Average annual total precipitation is 104 cm (National Climatic Data Center, 2022).

To understand how urbanization impacts tree community composition, we established field plots embedded in urban and rural natural parks (Fig. 1). The distribution of field plots spanned approximately 58 km from our eastern-most plot to our western-most plot and 39 km north to south. Natural parks were often irregularly shaped and various sizes ranging from 20 ha to 3000 ha.

2.2. Experimental design

We overlaid a 45 × 36 km grid over the greater Cleveland area, subdividing the area into 18 grid cells of 9 km by 12 km. Of those grid cells, three were eliminated because they did not contain a natural park, leaving 15 grid cells appropriate for study. Within each of these 15 grid cells, we established 2–9 research plots in each cell, with the number of research plots being relative to the size of the natural park.

A subset of 11 of these grid cells encompassed forested natural parks owned by Cleveland Metroparks and contained already-established long-term vegetation plots. For the grid cells containing these existing long-term vegetation plots, we first used data provided by the Cleveland Metroparks to determine the coordinates of forested long-term vegetation plots. We then selected our research plots for use by haphazardly choosing the coordinates of 10–15 forested long-term vegetation plots within relevant grid cells. Next, we used ground-truthing to determine if each long-term vegetation plot was appropriate for assessment, eliminating any of the long-term vegetation plots that were smaller than 20 × 25 m, obviously planted, or had an impenetrable woody shrub layer that prevented walking through the site. Plots were oriented north to south (the 25 m edge running N-S) where the topography allowed. If the topography did not allow for that orientation (e.g., if there was a steep slope or road) the plot was oriented with the long edge running east to west.

For grid cells without preexisting long-term vegetation plots, we

randomly placed plots within natural parks. The number of points was relative to the size of the natural park. We visited the coordinates of each randomly dropped point to determine if it was appropriate for assessment based on the same criteria listed above for the cells containing long-term vegetation plots. This resulted in 2–5 research plots per grid cell and a total of 59 research plots. Using this process, the rural natural parks were under-represented in our data set. To solve this problem, we added four plots to another rural natural park in 2020 using the same method described above. This resulted in a total of 63 plots in the study.

2.3. Classification of plots within natural parks as urban or rural

We used the 2010 US census block data to classify the land type of our study plots within each natural park as either urban or rural (US Census Bureau). The Census considered blocks with a population of 2500 or higher to be urban areas while all others were considered rural. Using this classification method, some natural parks included a mix of both urban and rural plots (See [Supplementary Materials, Table S1](#)).

2.4. Midstory and canopy tree and tree seedling abundance

Within each plot, we counted all midstory and canopy trees to determine abundance. We defined midstory and canopy trees as those greater than 5 cm diameter at breast height (DBH), noting species and DBH for each individual. In the case of multi-stemmed trees that were split below 1.3 m (i.e., the height of DBH measurement), each stem greater than 5 cm was recorded as an individual. We surveyed the midstory and canopy trees for 58 of the plots from June to August of 2019, and an additional 5 plots in June of 2020.

We used five 1 m² quadrats within each plot to determine seedling abundance. Four of the quadrats were inset 5 m from the plot's four corners and one quadrat was placed in the center of the plot. We recorded the species identity of each individual seedling under 1 m in height. Fifty-eight plots were assessed for seedlings from June to August of 2020. One long-term vegetation plot in an urban location that was 12 × 25 m was included in the analysis of midstory and canopy trees but was not assessed for seedling abundance.

2.5. Data analyses

We used four analyses to examine the effect of urbanization on forest communities located in Natural Parks. 1) We used PERMANOVA to test for an effect of land type (i.e. urban versus rural plots) on species composition and we used non-metric multidimensional scaling (NMDS) to visualize tree community composition; 2) We used similarity percentage analysis (SIMPER) and further investigated species-specific patterns of abundance in urban and rural plots using post-hoc negative-binomial regression in the midstory and canopy as well as seedling layers, separately; 3) We used a negative-binomial regression to test for an effect of urban versus rural land type on total tree abundance in the midstory and canopy as well as seedling layers, separately, because of the ability of this test to handle the overdispersion in our data; 4) Finally, we tested whether the urban versus rural land types impacted the Shannon Index and species richness using a Kruskal-Wallis rank test and similarly, we examined patterns of among-plot diversity (i.e. beta-diversity) using a multivariate homogeneity of group dispersions test.

1) For our PERMANOVAs, land type (urban or rural) was set as the predictor, and we included natural park as a co-variate to account for autocorrelation of plots located within the same natural park. PERMANOVAs were performed on Bray-Curtis dissimilarity matrices individually calculated for: abundance of midstory and canopy trees, and abundance of seedlings. For all PERMANOVA tests we used the *adonis()* function in the *vegan* package for R (Oksanen et al., 2020). We used NMDS to visualize tree community composition followed by MetaMDS to calculate model stress. For seedlings, we removed one

plot that had zero seedlings and one species for which there was only one individual, prior to running the NMDS and PERMANOVA.

- 2) For our SIMPER test, we used the *simper()* function to identify species contributing significantly to Bray-Curtis dissimilarity between urban and rural land types. Species identified as contributing to 3 % or more of dissimilarity between land type were analyzed individually by conducting species-specific tests of abundance between urban versus rural land types.
- 3) For our negative-binomial regression, we used the *glm.nb()* function from the MASS package (Venables and Ripley, 2002) in R with abundance of each species, separately, as the response variable and the land type (urban versus rural) as the explanatory variable.
- 4) We calculated species richness as the total number of species per plot. We calculated the Shannon index using the *diversity()* function with “Shannon” option in *vegan*. Both richness and Shannon index had non-normal distributions based on a Shapiro-Wilks test, so we used a Kendall rank order nonparametric test. We calculated between-plot diversity with the *betadisper()* function in *vegan*.

All analyses were completed in R version 4.2.1 (R Core Team, 2022).

3. Results

3.1. Community composition of tree species

We recorded 35 species of trees in the midstory and canopy. Of those species, only two were non-natives: *Acer platanoides* and *Malus domestica*. Non-metric multidimensional scaling showed that midstory and canopy tree communities in rural land types were more closely grouped in ordination space than those of the urban land type, meaning that there was greater similarity among rural plots compared to urban plots (Fig. 2). PERMANOVA indicated that urbanization had a significant effect on the composition of midstory and canopy trees (Table 2a). SIMPER analysis indicated that certain species contributed more to the differences between urban and rural communities than other species. Species which contributed greater than 3 % to dissimilarity were: *Acer saccharum* (16 %), *Acer rubrum* (13 %), *Fagus grandifolia* (10 %), *Liriodendron tulipifera* (6 %), *Prunus serotina* (4 %), and *Quercus rubra* (3 %; Fig. S1). Abundance of *Quercus rubra* was three times higher in the urban

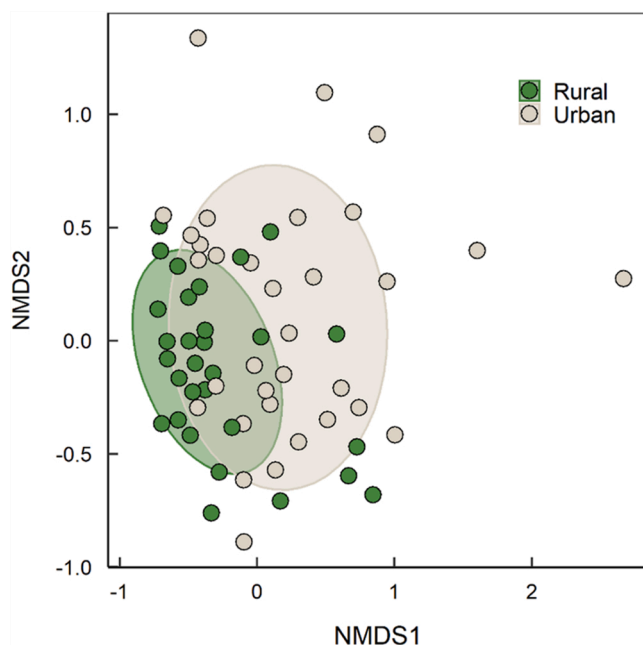


Fig. 2. NMDS showing differences in the midstory and canopy tree community composition from 63 plots in urban and rural natural parks. Stress= 0.21.

Table 2

PERMANOVA data table including F statistics for variables tested as explanatory terms for differences in abundance of species in (a) the midstory and canopy layer, and (b) the seedling layer.

Terms	df	F	p-value
(a)			
Urban/Rural	1	5.26	0.001
Natural Park	17	2.13	0.001
Residuals	44		
(b)			
Urban/Rural	1	3.40	0.002
Natural Park	16	1.28	0.046
Residuals	39		

land type compared to rural land type ($\chi^2 = 25.76$, $p < 0.001$) (Fig. 3). In contrast, *Fagus grandifolia* ($\chi^2 = 7.75$, $p < 0.01$) and *Liriodendron tulipifera* ($p < 0.001$) were 1.4 and 3.3 times more abundant in the rural land type, respectively (Fig. 3). Abundance of midstory and canopy *A. saccharum* ($\chi^2 = 1.68$, $p = 0.19$), *A. rubrum* ($\chi^2 = 0.5$, $p = 0.44$), and *P. serotina* ($\chi^2 = 2.1$, $p = 0.14$) were not significantly different between urban and rural land type.

Twenty unique taxa were recorded in the seedling layer (note *Ulmus*, *Carya*, and *Crataegus* were identified to the genus level). We found seedlings of the non-native species *Ailanthus altissima* and *Malus domestica*. Nonmetric multidimensional scaling showed rural land type forest communities clustered closer together relative to those of urban land type, indicating greater similarity among rural land type compared to urban land type (Fig. 3). PERMANOVA indicated that land type had a significant impact on community composition when accounting for natural park (Table 2b, Fig. 4). SIMPER analysis showed that the species contributing the most to these differences were *Acer saccharum* (22 %), *Acer rubrum* (14 %), *Fraxinus* spp. (14 %), *Prunus serotina* (7 %), *Liriodendron tulipifera* (6 %), *Quercus rubra* (6 %) and *Fagus grandifolia* (3 %; Fig. S2). Seedlings of *Prunus serotina* ($\chi^2 = 5.4$, $p = 0.02$) and *Quercus rubra* ($\chi^2 = 6.2$, $p = 0.01$) were twice and six times more abundant, respectively, in urban natural parks compared to rural natural parks. *Acer rubrum* ($\chi^2 = 15.66$, $p < 0.001$), *A. saccharum* ($\chi^2 = 4.07$, $p = 0.04$), and *Fagus grandifolia* ($\chi^2 = 15.2$, $p < 0.001$) were all more abundant in rural land type (Fig. 5). *Fagus grandifolia* was four times more abundant in rural compared to urban land type, while *Acer* spp. were two times more abundant in rural land type.

3.2. Diversity and Total Tree Cover

Richness and Shannon Index were not significantly different between land type for midstory and canopy trees (richness, $t = 2.6$, $p = 0.11$; Shannon, $\chi^2 = 1.7$, $p = 0.19$) or seedlings (richness, $t = 0.4$, $p = 0.5$; Shannon, $\chi^2 = 0.12$, $p = 0.72$). Total tree number in the midstory and canopy ($\chi^2 = 0.5$, $p = 0.5$) and seedling ($\chi^2 = 1.51$, $p = 0.22$) layers was not significantly different between urban and rural land types. There was a significant difference in beta-diversity for urban compared to rural land types, where urban natural parks had a higher level of variance compared to rural natural parks in the midstory and canopy layer ($F = 12.64$, $p < 0.001$) and in the seedling layer ($F = 7.99$, $p = 0.006$).

4. Discussion

We examined the abundance, diversity, and community composition of tree species in urban and rural natural parks in Cleveland, Ohio, to better understand how urbanization affects forest communities. Urban areas are typically considered disturbed, fragmented, and characterized by stressful abiotic conditions. Together, these factors are thought to constrain which species can survive and thrive in urban environments (Williams et al., 2009, Aronson et al., 2016). Below, we discuss evidence for the idea that urbanization can have unexpected effects on plant community composition, suggesting that the ecosystem services

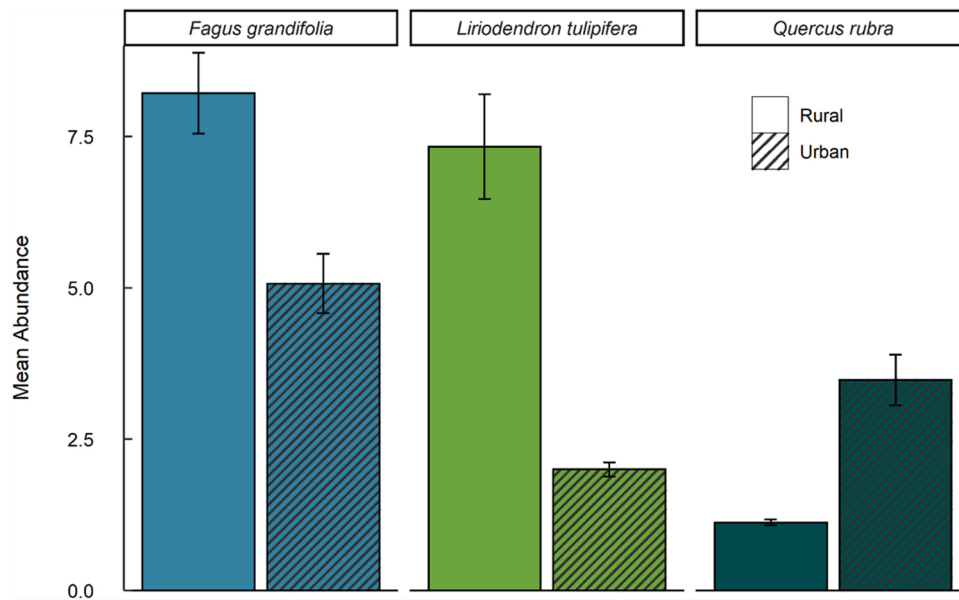


Fig. 3. : Mean abundance of commonly observed midstory and canopy tree species (based on SIMPER test) examined in urban and rural natural parks in and around Cleveland, OH. Bars indicate ± 1 standard error. Abundance was determined by the number of individual trees.

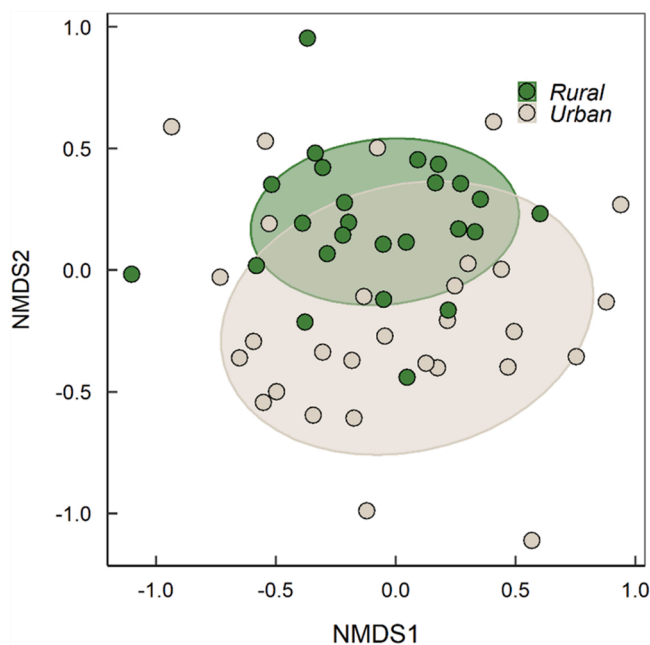


Fig. 4. NMDS showing compositional differences in the seedling layer community from 58 plots in urban and rural natural parks. Stress = 0.22.

provided, and the outcome of current management practices may be difficult to predict across natural parks within an urbanized region. We also provide evidence that more stress-tolerant tree species are more abundant in urban areas, but that other aspects of urbanization likely interact with stressful environmental conditions to shape the composition of urban tree communities in natural parks.

By comparing urban and rural natural parks, we can gain insight into how urbanization impacts forest plant communities. One concern in urban areas is the potential for increased prevalence of invasive and non-native tree species, and while we did encounter some of these species in the urban area, our study adds to previous work on the canopy layer showing that urban natural forests predominantly have native trees (Pregitzer et al., 2019, Piana et al., 2021, Doroski et al., 2022).

More interestingly, within this context in which most of the trees we encountered in the urban natural parks were native species, we still found distinctions between urban and rural areas in the composition of both midstory and canopy as well as seedling plant communities (Fig. 2&4). With this effect observed in native plants, this provides evidence that urbanization itself can alter the species composition of natural parks, and that over time, urban parks may become even more distinguishable from their rural counterparts.

The potential for stressful environmental conditions created by urbanization to alter the forest community is supported by our observation that *Quercus rubra*, a light-loving species considered well-equipped to tolerate stressful temperature conditions and drought (Arend and Scholz, 1969), was more abundant in urban natural parks. This may suggest that urbanization favors a more stress resistant plant community, or this pattern could be driven by *Q. rubra* in rural natural areas being outcompeted by more shade-tolerant trees such as *Fagus grandifolia* (American beech) and *Acer saccharum* (Sugar Maple) (Peters et al., 2020). In support of this idea, we found that although *Q. rubra* trees are more abundant in the urban plots, their average diameter in the rural forests was significantly larger (Fig. S3). Thus, our study suggests that the rural forests we examined may be at a later successional stage than the urban forests, perhaps having had more time to establish since disturbance.

While our study does not allow us to disentangle the mechanisms that led to the compositional differences we observed, our work highlights the pitfalls of predicting the effect of urbanization based solely on established trees. Despite this, studies on urban seedling recruitment remain rare (Piana et al., 2021, Doroski et al., 2022). Here, we found points of both agreement and disagreement between midstory and canopy versus seedling layers. Consistent with our findings in midstory and canopy trees, we found that the seedling layer was made up of predominantly native species in both urban and rural natural parks. The species composition of the seedling layer within the urban forest, however, was characterized by higher abundance of *Quercus rubra* seedlings relative to rural forests (Fig. 5). With this pattern being observed both for seedlings as well as midstory and canopy trees, this provides stronger evidence that some feature particular to *Q. rubra* may allow this stress resistant species to incur an advantage in the urban environment over other tree species.

Our data also suggests that *Prunus serotina* may experience an

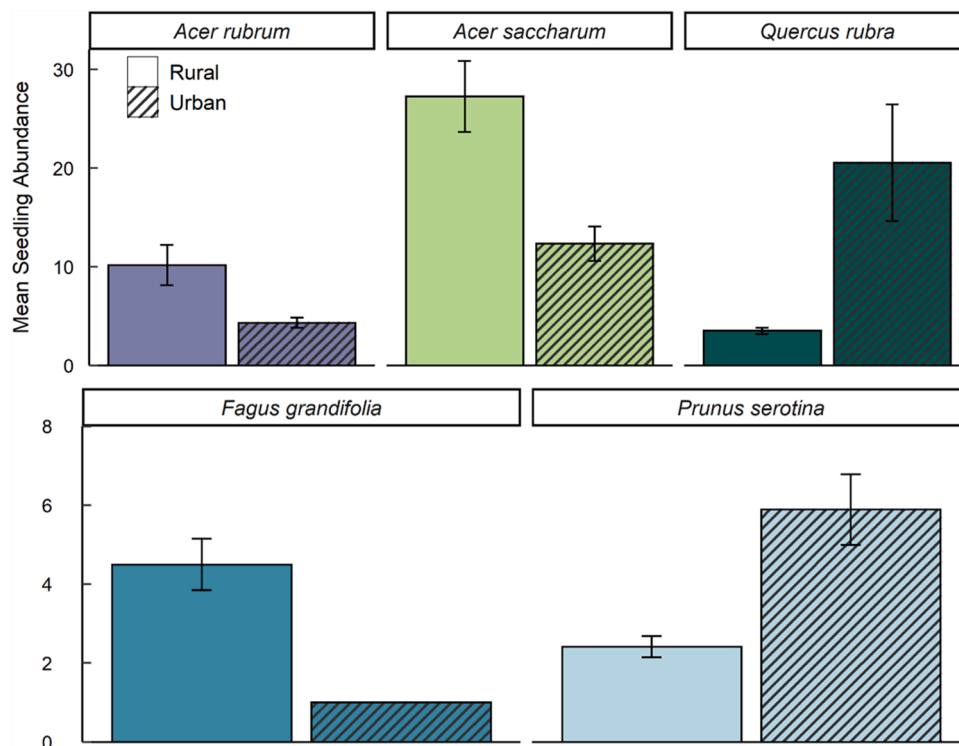


Fig. 5. Mean abundance of commonly observed tree seedling species (based on SIMPER test) examined in urban and rural natural parks in and around Cleveland, OH. Bars indicate ± 1 standard error. Abundance was determined by the number of individual trees.

advantage over other native tree species in the urban environment. This species was more abundant in the seedling layer of urban natural parks compared to rural areas. As with *Q. rubra*, we also saw that most rural parks had much lower abundance of *P. serotina* in the midstory and canopy layers compared to urban sites, but two rural plots had high abundance of *P. serotina* (Table S1) and these trees were relatively large (Table S3, Fig. S4). This high variation in abundance across plots within rural parks resulted in no significant difference for midstory and canopy *P. serotina* abundance across urban and rural land type, however, these data do add further evidence to the idea that both the stress tolerance of trees and the successional stage of the forest is not the same across urban and rural locations. Interestingly, *Prunus serotina* has a wide distribution and is expected to expand its range under warming climate models, which could point to higher performance of tree species with greater thermal tolerance in urban areas (Segura et al., 2018).

In contrast, we saw lower abundance of *Acer* spp. in the seedling layer of urban areas, despite *Acer* spp. making up a substantial component of nearby rural forests. *Acer saccharum* is considered less stress-tolerant compared to *Q. rubra* and *P. serotina*, providing another piece of evidence that more stress tolerant tree species are being favored in the urban environment. *Acer rubra* is also considered a highly stress-tolerant pioneering species, which could suggest that it would be more prevalent in urban areas in our study, but other features may come into play in determining the abundance of species in urban forests. For example, we found that *Fraxinus* spp. were uncommon among midstory and canopy trees in both forest types, likely due to past infestations of emerald ash borer, which host on *Fraxinus* spp. trees (Poland and McCullough, 2006). Yet, *Fraxinus* spp. made up a significant component of the seedling layer in both urban and rural locations. While this may suggest high suitability of the urban areas for *Fraxinus* recruitment, ultimately these *Fraxinus* seedlings may succumb to future emerald ash borer infestations, and thus never represent a large component of the urban forest of the future. Taken together this highlights the complexity of interpreting urbanization effects on plant communities. While our study design does not allow us to distinguish between these nuanced possibilities of environmental

stress versus successional causes for the patterns we observed, our study does suggest future work is needed examining the relative fitness of these native seedling species in both urban and rural environmental contexts.

4.1. Biodiversity

Given the increased prevalence of stressful environmental conditions in urban areas, one hypothesized effect of urbanization is homogenization of the biotic environment, which can be influenced by multiple drivers. Urban areas tend to have decreased ecosystem complexity compared to than do more rural environments as a function of occupying space within the human-built environment (McDonnell et al., 1997). Further, natural areas within these urban spaces often tend to attract and support a subset of species that thrive in edge habitat (McKinney, 2006). Additionally, homogenization may come about due to significant invasion in disturbed areas (McKinney, 2006). In our study, we did find three non-native tree species, *Acer platanoides*, *Malus domestica*, and *Ailanthus altissima*, but these were relatively low in abundance, and so had little influence in our analysis. In addition, we found that diversity was the same at both urban and rural natural parks within the midstory and canopy as well as the seedling layers. Other studies have shown that biodiversity can decrease in urban forests despite a lack of invasives (Kolbe et al., 2016). Interestingly, we did find significantly higher beta-diversity in urban compared to rural natural parks, and this was true in both the midstory and canopy layer as well as the seedling layer, i.e., there is greater variation in community composition among urban research plots than among rural plots. This result contrasts with large-scale studies reporting low urban diversity due to homogenization of plant types (McKinney, 2006, Groffman et al., 2014) and regional studies that have found little to no impact of urbanization on beta-diversity (Blood et al., 2016, Templeton et al., 2019).

Factors contributing to the higher beta-diversity across urban natural parks likely include both pre-existing topographical variations combined with small-scale variation in land-use history resulting in

fragmentation of the urban landscape (Swan et al., 2011). This may result in somewhat unique “islands” of habitat scattered across the urban landscape. The natural parks sampled herein, both urban and rural, have been opportunistically preserved and converted into conserved and recreational parklands. In fact, while natural parks are defined in part by the fact that they have never been cleared for urbanization, they undoubtedly have complex land use history including forestry and agriculture, and these effects can persist for decades or longer (Foster et al., 2010). Interestingly, natural parks within the heart of Cleveland are disproportionately restricted to a small buffer of land surrounding bodies of water, which may come about because riparian zones are viewed as more charismatic and important components of the landscape, leading to their preferential preservation (Ryan et al., 2010). For example, one of our urban natural parks, the Ohio and Erie Canal Reservation (Fig. 1, Table 1) has a relatively narrow amount of wooded land surrounding an historic canal that runs through the greater Cleveland Area. This idea that riparian zones are disproportionately represented in urban compared to rural natural areas is supported by the fact that, in our study, species associated with riparian zones were only captured in urban plots (e.g., *Populus deltoides* and *Acer negundo*). In addition, riparian zones have been a focus of restoration in recent years which may lead to their over-representation in urban natural areas (Flinn et al., 2018). Thus, the results of our study suggest that it may be unlikely that species composition of urban natural parks would be the same as the overall plant community of surrounding rural areas and emphasizes that future work is needed to unravel the numerous factors related to urbanization (e.g., invasive species, fragmentation, environmental conditions) and their complex interactive effects on patterns of biodiversity.

Intriguingly, riparian zones in urban areas are thought to be important corridors for dispersal and may be the only natural connection between urban forest patches. Thus, indirect changes in hydrological processes caused by urbanization could change the connectivity of urban riparian zones, which in turn may alter the dispersal of plant species (Brice et al., 2016). Tree species that have lower seed production or limited dispersal may, therefore, be underrepresented in these forests (Bonte et al., 2012), providing a potential mechanism for greater abundance of bird-dispersed species in urban forests (Olejniczak et al., 2018). In fact, *Prunus serotina*, one of the most abundant species in the urban areas in this study, is bird dispersed (Marquis, 1990), which may provide a dispersal advantage over other species, leading to over-representation of this dispersal type in fragmented urban landscapes (Doroski et al., 2018). Interestingly, roads and paved areas may present a barrier to species which are dispersed by small mammals (Niu et al., 2018), which is in direct contrast to our finding that *Q. rubra* was more abundant in urban versus rural plots. This points to the idea that complex factors likely interact to produce the patterns of plant community composition in urban remnant forests (Ramalho et al., 2014), which is ultimately supported by our finding of high variation in the community composition across urban plots.

5. Conclusions

The natural urban forests of northeast Ohio, known colloquially as the “Emerald Necklace”, are an invaluable component of the greater urban forest. As land managers seek to increase the forested canopy of the city by 30 % by 2040 (Cleveland Tree, 2018), maintaining the health of existing urban forest resources will be critical to reach this goal. We showed that stress tolerance is likely to be a key feature that imparts success in the urban environment, pointing directly to the urban heat island effect as critical mechanisms shaping the urban tree community. We also found evidence that the successional stage differs across urban and rural natural parks, and that dispersal of tree seedlings may be a driving factor in shaping the tree communities of urban natural parks. The observational nature of the study makes it impossible for us to determine which of these mechanisms that may be driving the

compositional differences we observed, but our study emphasizes that widely held assumptions about the urban forests are not universally true across urban sites, and successfully managing these important natural resources will require local characterization of the composition of all strata of forests. Our finding of compositional differences in the seedling layer highlight that the future urban forest may be moving towards one composed of stress-tolerant species like *Q. rubra* and *P. serotina*. Further, our finding that urban natural parks harbor native tree diversity emphasizes the importance of conserving them. Lastly, we show that patterns of forest diversity across urban environments will be difficult to predict, but high beta-diversity could contribute to forest resiliency (Dell et al., 2019).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ufug.2023.127864.

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